

Model-Based Software Design, A.Y. 2024/25

Laboratory 2 Report

Components of the working group (max 2 people)

- Name Surname, student ID
- Federica D'Ugo, 345772
- Francesca Fornasier, 338688

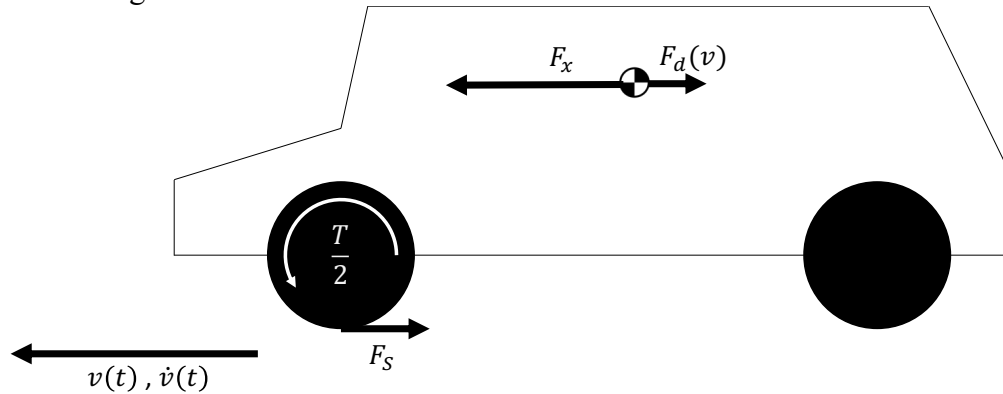
External interfaces of the plant

Name	Direction	Type
Requested_Torque_Nm	Input	CAN
Vehicle_Speed_kmh	Output	CAN
Automatic_Transmission_Selector	Input (from the driver to the controller)	CAN {P, R, N, D, P}
Selected mode/errors	Output (to the driver)	CAN

Equations of the plant

The plant considered in this model is the so-called *Vehicle Longitudinal Dynamics*.

Considering:



- $\dot{v}(t)$ the vehicle acceleration, expressed in $[m/s^2]$
- $v(t)$ the vehicle longitudinal speed, expressed in $[m/s]$
- m the vehicle mass, expressed in $[kg]$
- $F_x(t)$ the longitudinal force applied to the vehicle center of gravity, expressed in $[N]$
- $F_s(t)$ the longitudinal force applied to the wheel on the terrain, expressed in $[N]$
- $F_d(v)$ the longitudinal force applied to the vehicle center of gravity due to the frictions with air and terrain, expressed in $[N]$
- I the moment of inertia of each one of the wheels, expressed in $[kg \cdot m^2]$
- r the radius of the wheel, expressed in $[m]$
- $\omega(t)$ is the angular speed of the wheel, expressed in $[rad/s]$
- $\omega_e(t)$ is the angular speed of the engine/electrical motor, expressed in $[rad/s]$
- $\omega_{eRPM}(t) = \frac{\omega_e(t) \cdot 60}{2\pi}$ is the angular speed of the engine/ electrical motor, expressed in [revolutions per minutes]
- $\dot{\omega}(t)$ is the angular acceleration of the wheel, expressed in $[rad/s^2]$
- S is the frontal surface of the car, expressed in $[m^2]$
- $c_x = 0.3$ is the automobile drag coefficient
- ρ is the average density of air at sea level in standard conditions $\rightarrow \rho = 1.25 \text{ kg/m}^3$
- i_g is the gearbox reduction ratio
- i_f is the final drive reduction ratio
- $i_t = i_g \cdot i_f$ is the total power train reduction ratio.

An extremely simplified model can be obtained as follow:

$$\dot{v}(t) = \frac{F_x(t) - F_d(v)}{m} \quad (1)$$

where $\dot{v}(t)$ is the vehicle acceleration, m is its mass, $F_x(t)$ is the longitudinal force applied to its center of gravity by the effects of the torque applied on the wheels, and $F_d(v)$ is the sum of the friction forces on the vehicle due to wheel-terrain and vehicle-air interactions.

Considering that the torque is equally split between the two wheels (valid only on straight tracks)

$$T(t) - 2 \cdot F_s(t) \cdot r = 2I \cdot \dot{\omega}(t) \quad (2)$$

the absence of slipping:

$$\begin{cases} 2 \cdot F_s(t) \cdot r = F_x(t) \\ v(t) = 2 \pi \cdot r \cdot \omega(t) \end{cases}$$

and considering the moment of inertia of the wheels $I = 0$, we can define the following equation, given that $T(t) = 2 \cdot F_s(t) \cdot r \rightarrow F_s(t) = \frac{T(t)}{2 \cdot r}$.

The drag force that limits the maximum speed of the vehicle is equal to:

$$F_d(v(t)) = X_{air} \cdot (v(t))^2 + X_{tyres} \cdot v(t) \quad (3)$$

where:

$$X_{air} = \frac{1}{2} \cdot S \cdot c_x \cdot \rho \quad (4)$$

and, as usually modeled:

$$X_{tyres}|_{X_{tyres}(50 \frac{\text{km}}{\text{h}}) = X_{air}(50 \frac{\text{km}}{\text{h}})} \rightarrow X_{tyres} = \frac{X_{air} \cdot 50}{3.6} \quad (5)$$

By substituting the (2) equation in (1), and by integrating both sides, we obtain:

$$\begin{aligned} v(t) &= \frac{1}{m} \int_0^t F_x(t) - F_d(v(t)) dt = \frac{1}{m} \int_0^t 2 \cdot F_s(t) - F_d(v(t)) dt = \\ &= \frac{1}{m} \int_0^t \frac{T(t)}{r} - F_d(v(t)) dt \quad (6) \end{aligned}$$

and, by substituting (3) in (6):

$$v(t) = \frac{1}{m} \int_0^t \frac{T(t)}{r} - X_{air} \cdot (v(t))^2 - X_{tyres} \cdot v(t) dt \quad (7)$$

Remember that the integrator block of Simulink requires an initial condition corresponding to the vehicle's longitudinal speed at the beginning of the simulation, $v(0)$. A possible configuration of the integration block is shown in Figure 2.

During the model development, put all the needed gain to obtain as an output of the physical model a speed expressed in km/h.

To simulate the slope θ of the terrain, it is possible to add the gravity force $F_g(\theta)$ as follows:

$$v(t) = \frac{1}{m} \int_0^t \frac{T(t)}{r} - X_{air} \cdot (v(t))^2 - X_{tyres} \cdot |v(t)| \cdot dt + m \cdot g \cdot \sin(\theta) \quad (8)$$

With $g = 9.81$ the gravity acceleration on Earth.

Reasonable values for an electric compact car can be:

- $m = 1600$ kg
- $r = 0.3$ m
- The torque T (at the wheel) can vary in the range $[-60; 960]$ Nm
- $S = 3.5$ m²
- $c_x = 0.3$

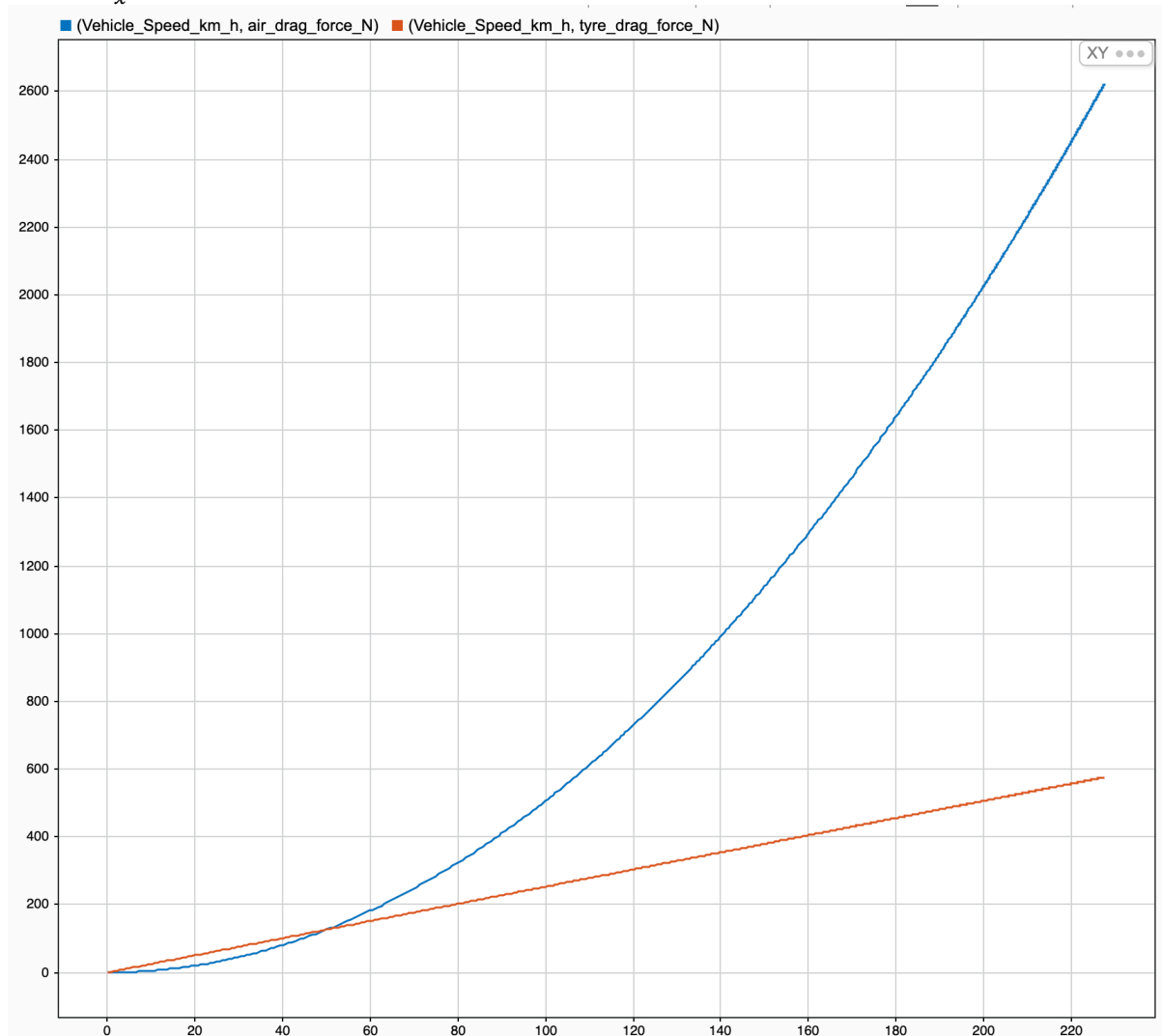


Figure 1 Graph showing drag forces of tires (in orange) and air (in blue) at various speeds. It is possible to observe that, as imposed in equation (5), $X_{tyre} = X_{air}$ at 50 km/h. Below this speed, the tire drag is dominant, after that, the air drag is dominant. Moreover, it is possible to see the top speed of the car (around 230 km/h) when $F_x = \frac{T}{r} = X_{tyre} + X_{air}$, with $F_x = 3200$ N.

With those values, the top speed on level ground reachable by the car is about 230 km/h, where the drag forces equal the traction force (3200 N).

Considering the reverse direction, the maximum speed reachable with a limitation of -60 Nm is about 45 km/h.

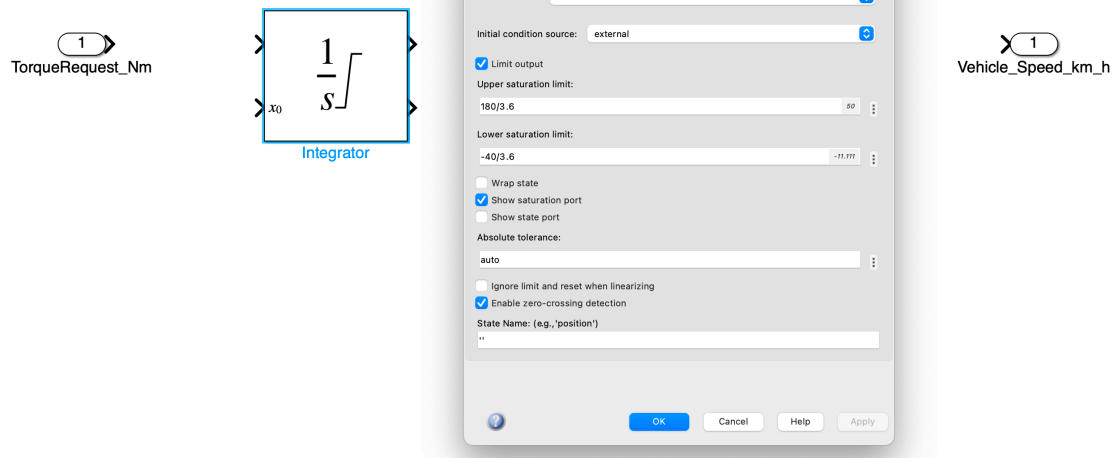


Figure 2 Settings window for the Integrator block of Simulink

Use these values (with a certain tolerance, for example, 10 %) to saturate the integrator block.

To make the model more realistic, it is possible to compute the torque request at the engine/motor. A typical ratio value for transmission of an electric car with a single gear can be around $i_t = 12^1$.

All the initialization parameters of the model are automatically loaded model by a callback of the function `init_fn` as shown in Figure 3.

Usually, the first gear of a car has a transmission ratio about $i_g = 3$, hence the engine makes 3 complete rotations very single rotation of the pinion gear of the differential. In the differential (final drive), the pinion gear makes about $i_f = 4$ revolutions for every revolution of the axle shafts. From here, the typical ratio of $i_t = 12$ between the torque at the wheel and the torque at the engine/electrical motor.

¹ Usually, the first gear of a car has a transmission ratio about $i_g = 3$, hence the engine makes 3 complete rotations very single rotation of the pinion gear of the differential. In the differential (final drive), the pinion gear makes about $i_f = 4$ revolutions for every revolution of the axle shafts. From here, the typical ratio of $i_t = 12$ between the torque at the wheel and the torque at the engine/electrical motor.

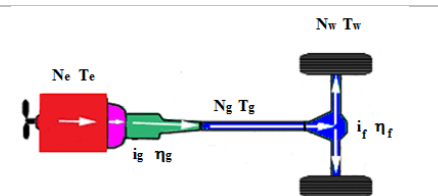


Figure from http://www.thecartech.com/subjects/auto_eng/Gearbox_ratios3.htm

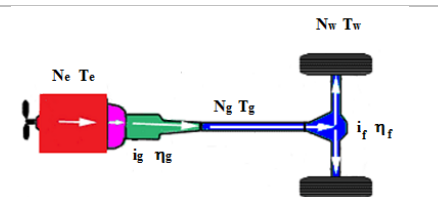


Figure from http://www.thecartech.com/subjects/auto_eng/Gearbox_ratios3.htm

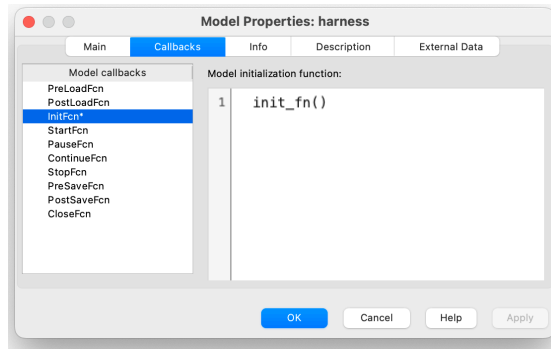
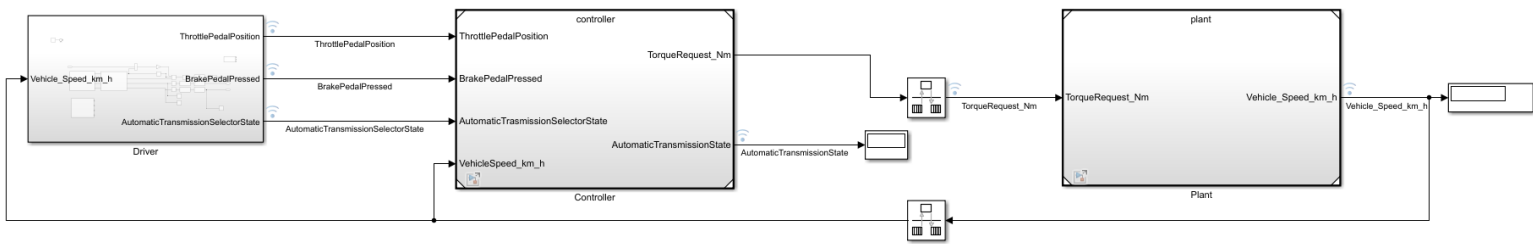


Figure 3 *init_fn* callback configuration in the harness model properties.

Description of the whole system

Draw the I/O block diagram of the plant and the controller, showing how they interact.



The Controller receives inputs from the driver, including ThrottlePedalPosition, BrakePedalPressed, and AutomaticTransmissionSelectorState. Additionally, it receives the VehicleSpeed_kmh from the Plant, which changes based on the TorqueRequest_Nm calculated by the Controller. In addition to TorqueRequest_Nm, the Controller produces another output, the AutomaticTransmissionState.

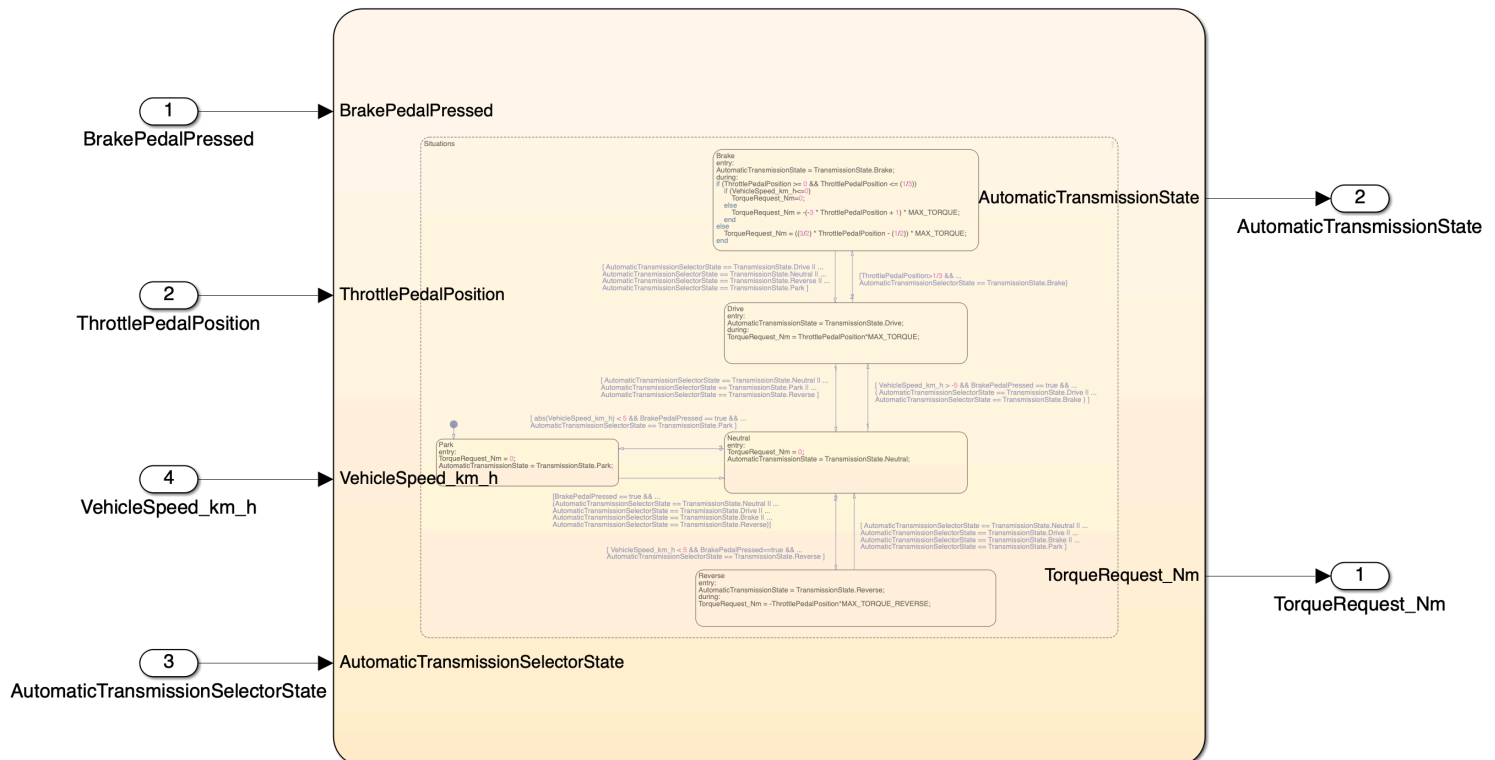
Controller SW Unit specifications

Provide a brief description of the Controller functionalities and its interfaces.

Interfaces

Name	Unit	Type ²	Data Type ³	Dimension	Min	Max
BrakePedalPressed	inherit	input	boolean	-1	[]	[]
ThrottlePedalPosition	inherit	input	single	-1	[]	[]
AutomaticTransmissionSelectorState	inherit	input	Enum: TransmissionState	-1	[]	[]
VehicleSpeed_kmh	km/h	input	single	-1	[]	[]
TorqueRequest_Nm	N*m	output	single	-1	[]	[]
AutomaticTransmissionState	inherit	output	Enum: TransmissionState	-1	[]	[]

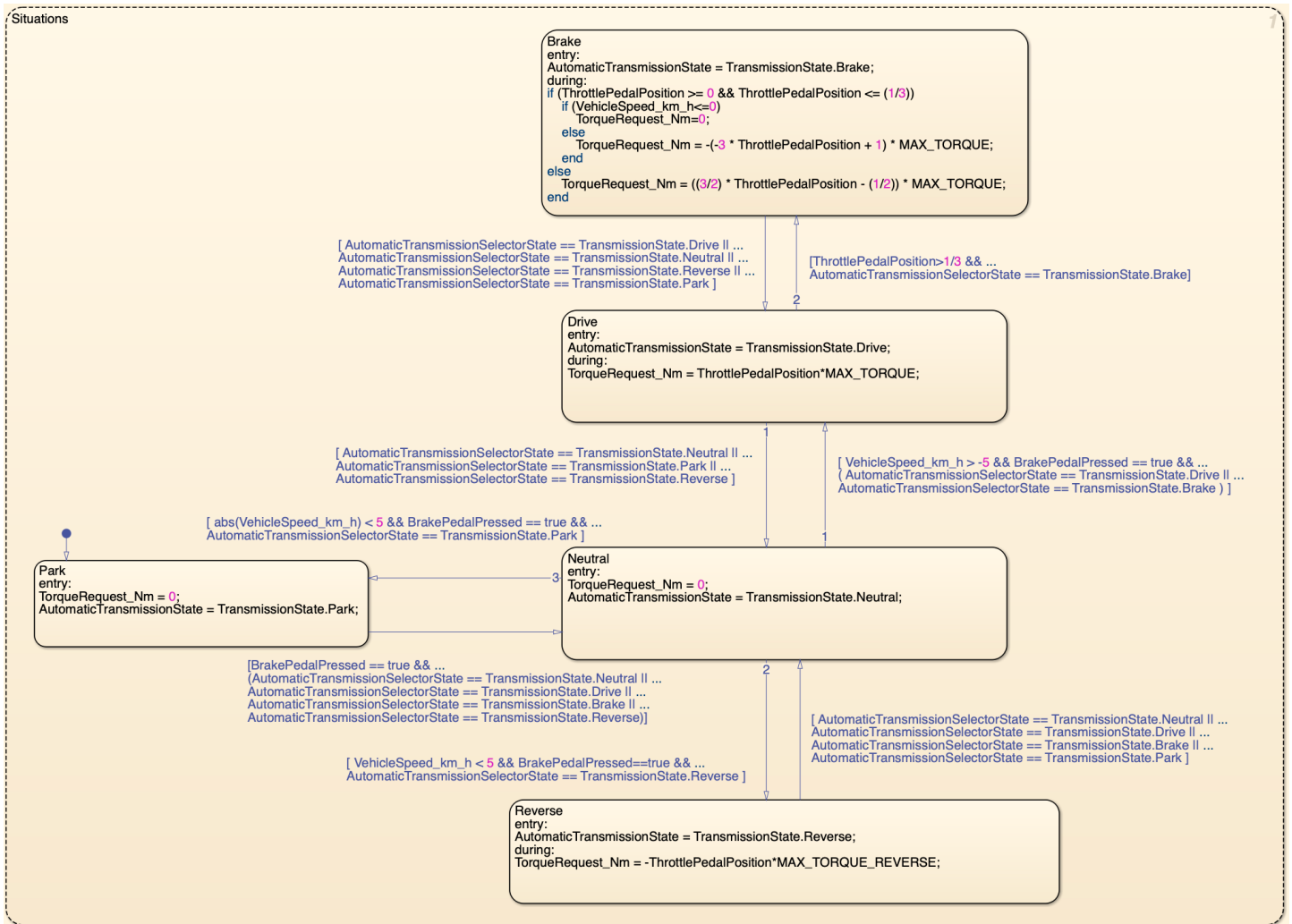
Draw the Finite State Machine (FSM) representing the controller logic



² Input, Output, Local, Global, Volatile

³ Struct, Double, Integer, Enum, Boolean, etc...

Comment on the design choices of the FSM, which are not trivial to be understood just by analyzing the controller logic.



Each driving mode is represented by a distinct state block of the finite state machine, and we modelled the transitions between them imposing the correct conditions defined in the design specification. Upon entering each state, a specific entry action is executed to update the AutomaticTransmissionState variable with the value corresponding to the current driving mode.

In Park state, which also serves as the starting point of the finite state machine, we have an additional action to set the torque to zero ensuring the vehicle remains still.

The same action is performed in the Neutral state, where the wheels are completely disengaged from the motor. As a result, even though the engine is still running, no power is transmitted.

In the remaining three states (Reverse, Drive and Brake), torque is actively managed based on the throttle pedal position.

- In Reverse, the torque is negative and proportional to the pedal position: TorqueRequest_Nm = -ThrottlePedalPosition*MAX_TORQUE_REVERSE
- In Drive, the torque is positive and similarly computed as: TorqueRequest_Nm = ThrottlePedalPosition*MAX_TORQUE
- In Brake we have to distinguish between acceleration and regenerative braking: If the pedal position is greater than 1/3 the vehicle is in the acceleration region. The

torque is positive and proportional to how far the pedal is pressed beyond this point, scaled such that the maximum torque request is obtained when the pedal is fully pressed and 0 torque is imposed at the neutral point. If the pedal position is below 1/3 the vehicle is in the regenerative braking region, a negative torque is applied, reaching its maximum at full release (position = 0) and decreasing linearly to zero at the neutral point. In regenerative braking region it is also necessary to check whether the vehicle's velocity is zero and if it is, further negative torque must be avoided to prevent the car from moving backward unintentionally.

In addition, the transition from one state to another must be allowed not only when the destination state corresponds to the state selected by the driver, but also when it represents an intermediate state necessary to reach the desired one (e.g. transition through Neutral to go from Reverse to Drive or Brake). All these additional conditions are specified as transition conditions.

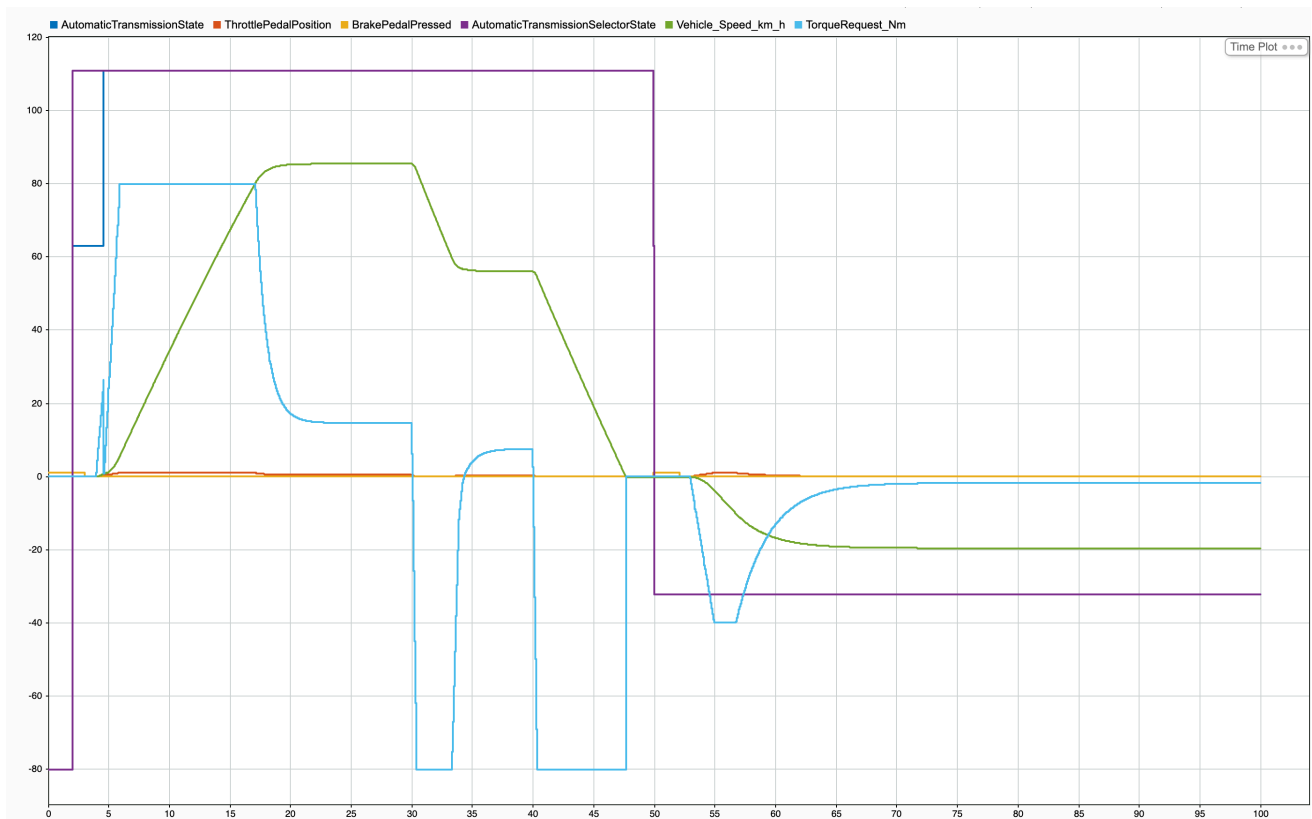
Transition conditions:

During all the transitions we have to check the AutomaticTransmissionSelectorState, to define which driving mode we want to reach

- If the brake pedal is pressed we can pass directly from Park to Neutral, this condition is also used to switch to other modes, like Drive, Brake and Reverse;
- From Neutral to Park and to Reverse it was necessary to have the vehicle speed lower than 5 and the brake pedal pressed
- From Reverse to Neutral, from Brake to Drive and from Drive to Neutral, there are no conditions other than the selected mode chosen from the driver
- From Neutral to Drive we had to have the vehicle speed higher than -5 and the brake pedal pressed. This condition is also used to switch to Brake
- From Drive to Break the throttle pedal position must be greater than 1/3

Comment with plots of the results obtained from the test cases (it is suggested to use the Data Inspector)

- *Plot for test pattern 1:*



As shown in the plot, the controller successfully produces the desired behavior.

The AutomaticTransmissionSelectorState begins in the Park state, which is the initial state of the finite state machine. In this driving mode the torque request is zero and, since the car is still, also the speed is zero.

After 2.3 seconds, the AutomaticTransmissionSelectorState transitions to Brake. Since the brake pedal is pressed and the speed is above -5 km/h, it transitions through Neutral and proceeds to Drive, where it temporarily remains. Once the throttle pedal exceeds one-third of its full travel, the transition condition is met and the state changes to Brake.

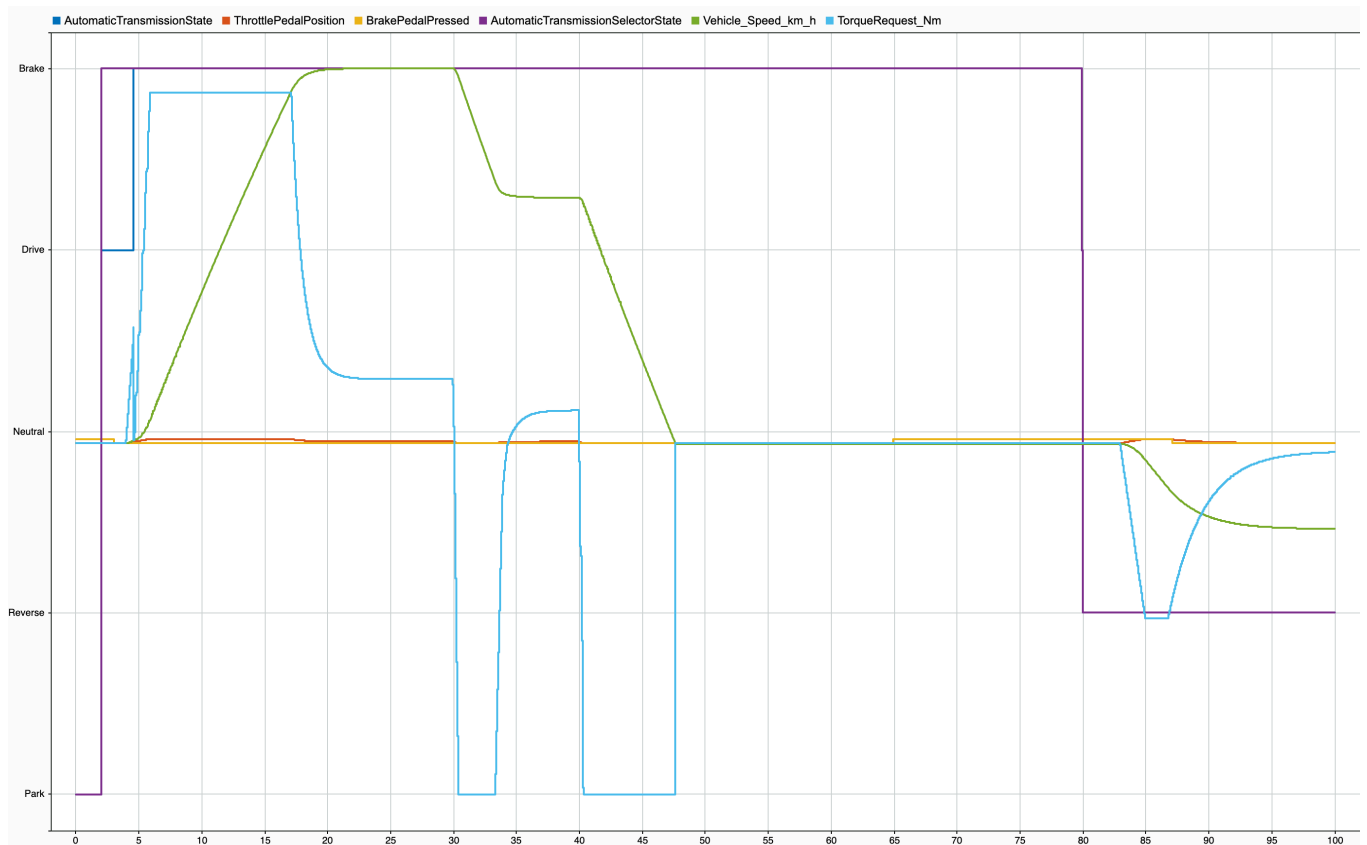
The ThrottlePedalPosition then reaches its maximum value (1), resulting in the maximum torque request of 80 Nm, which causes the vehicle speed to increase. Around 17 seconds, the throttle position decreases to approximately 0.5. Although the torque request remains positive, it is significantly reduced compared to the previous value, and the vehicle speed remains nearly constant.

At 30 seconds, the throttle pedal is fully released, leading to the application of the maximum negative torque. It is then briefly pressed again to a position slightly above the neutral point before being released once more. Consequently, the vehicle speed decreases, stabilizes momentarily, and then decreases again with the steepest slope. This trend continues until 47.5 seconds, when the speed reaches zero. At this point, the controller imposes zero torque to prevent the vehicle from rolling backward.

At 50 seconds, the AutomaticTransmissionSelectorState transitions to Reverse, followed immediately by the AutomaticTransmissionState, as the condition on vehicle speed is met and the brake pedal is pressed.

At 53 seconds, the throttle pedal begins to be pressed again, increasing to its maximum position before gradually decreasing and settling at a small value slightly above zero. As a result, an increasing negative torque is initially requested, reaching its maximum of -40 Nm, and then gradually decreasing to approximately -1.7 Nm. The vehicle's speed, which starts at zero, increases in absolute value (accelerating more rapidly when the torque is high and more slowly as the torque diminishes) eventually stabilizing around -19.5 km/h.

- *Plot for test pattern 2:*



As in the previous case, the AutomaticTransmissionSelectorState starts in Park and transitions to Brake. The AutomaticTransmissionState passes through Neutral and temporarily remains in Drive until the condition $\text{ThrottlePedalPosition} > 1/3$ is satisfied, after which it transitions to Brake mode.

The throttle pedal position increases from 0 to 1 (fully pressed), resulting in the application of the maximum torque request (80 Nm) and a corresponding increase in vehicle speed with the steepest slope. At approximately 17 seconds, the ThrottlePedalPosition decreases to 0.5, which stabilizes the speed around 85.5 km/h.

Between 30 and 33 seconds, the throttle is fully released, resulting in the application of the maximum negative torque, which causes the speed to decrease. Shortly afterward, the pedal is briefly pressed to about 0.4; the torque becomes positive but lower than before, and the speed remains nearly constant with a slight decrease. The pedal is then released again, reaching a position of 0, which results in the application of the maximum negative torque until the vehicle speed drops to zero. At this point, to prevent

any unintended backward motion, the controller sets the torque to zero, and the vehicle remains stationary.

At 80 seconds, the AutomaticTransmissionSelectorState transitions to Reverse, and the AutomaticTransmissionState follows immediately, as the condition requiring the brake pedal to be engaged is already satisfied, having been continuously pressed from 65 to 87 seconds.

At 83 seconds, the throttle pedal begins to be pressed, eventually reaching its maximum position before being slowly released. As a result, the torque request becomes negative, increasing in magnitude up to the maximum (-40 Nm) and then decreasing more gradually. This leads to a vehicle speed in the reverse direction that initially increases rapidly and then more slowly as the torque diminishes.